RaphCare feature checklist

PDF companion: `raphcare-feature-checklist.pdf` (regenerate with `scripts/Export-RaphCareFeatureChecklistPdf.ps1` whenever this file changes).

Plain-language twin for non-technical partners: `raphcare-feature-checklist-partner.md` (and its PDF). Keep both in sync when status changes.

Track progress across **API**, **Web admin panel**, and **Mobile** (patient app). Design / visual parity for Mobile stays in `../Mobile_Concept_Port.md`; release gates stay in `Mobile_Release_Ready_Checklist.md`.

Overall completion (this checklist): 78%
Without wearable Phase 2+ metrics (activity / sleep / stress / temp / ECG / glucose rows): **82%**

How to mark items and score %

- `[x]` done = 100% of that item
- `[x] ... (partial)` = 50%
- `[ ] ... (partial)` = 25% (started, mostly open)
- `[ ]` not done = 0%
- `(blocked: ...)` still scored (usually 0% or 25% if also partial)
- `(out of scope)` and open `(optional)` / `(optional / later)` items are **excluded** from section and overall %
- Vertical table rows (API / Client / Mobile UI) count as one item = average of the three cells
- Section % = average of that section's scored items
- Overall % = average of all scored items in Steps 1-4 plus Cross-cutting

Each step ends with a short status note and its section %. Recalculate when you change checkboxes.

Layer order for new HTTP contracts

Backend (API + Application + Persistence) -> `RaphCare.Client` -> Web / Mobile. Do not duplicate API contracts inside Mobile.

Last reviewed: 2026-08-23

Step 1: Admin clinic / hospital ops (outpatient) - 100%

Work under `api/admin/clinics` -> `IAdminClinicService` -> Blazor `Pages/Admin/Hospitals/*`.

API + Client

- [x] Clinic list / detail / update
- [x] Register clinic + claim by registration number + ensure membership

- [x] Patients: list, detail, grant / revoke access
- [x] Staff: invite, role, resend, cancel pending, remove
- [x] Facilities CRUD (physical + virtual)
- [x] Dashboard metrics
- [x] Providers: list, detail, create, set active, schedules create / delete
- [x] Appointments: list, book, cancel, reschedule
- [x] Visits: start, get, complete, record vitals
- [x] Clinic devices list
- [x] Tele-session start (Agora join info for admin)

Admin panel (Web)

- [x] Hospitals index / register / claim
- [x] Hospital detail (overview, facilities, patients, providers, staff)
- [x] Patient detail page
- [x] Provider detail + schedules
- [x] Appointments page
- [x] Visit page + vitals
- [x] Tele join page
- [x] Admin dashboard
- [x] Web UI language switcher (en / fr / ln / sw)
- [ ] Phone-usable admin (responsive layout, then thin install) `(optional / later)`: same web portal on phones; not a clinician MAUI app; not a full PWA. See `../15_Web_Admin_On_Phone.md`

Mobile

- [ ] Admin / clinician hospital ops `(out of scope)`: patient app only; admin stays on Web

Step 1 status

100%. Outpatient hospital admin is end-to-end on API + Client + Web. Clinician Mobile stays out of scope. Phone use of the same web admin is later (optional); see `../15_Web_Admin_On_Phone.md`.

Step 2: Inpatient (wards / beds / admissions), MVP - 100%

New vertical: Domain `Ward` / `Room` / `Bed` / `InpatientAdmission`, migration `AddInpatientBedsAndAdmissions`, `AdminClinicsController` inpatient actions, Client methods, Web `Inpatient.razor`.

API + Application + Persistence

- [x] Domain entities + EF configs + migration
- [x] `GET {clinicId}/inpatient/board`

- [x] `POST {clinicId}/wards` / `rooms` / `beds`
- [x] `POST {clinicId}/admissions` (admit; one active stay; bed Available -> Occupied)
- [x] `POST {clinicId}/admissions/{id}/discharge` (free bed)
- [x] `IAdminClinicInpatientQueryService` board aggregation
- [x] Update / deactivate / delete ward, room, bed
- [x] Set bed status to `Maintenance` (status exists on domain; unused in handlers)
- [x] Transfer patient between beds
- [x] Admission history / get-by-id
- [x] Soft-delete or archive capacity
- [x] Demo seed sample wards / beds `(optional)`

Client

- [x] `GetInpatientBoardAsync`, `CreateWard/Room/BedAsync`, `AdmitPatientAsync`, `DischargeAdmissionAsync`
- [x] Models in `ClinicOperationalModels` (`ClinicInpatientBoard`, ward/room/bed/admission)
- [x] Client methods for update / maintenance / transfer / admission history

Admin panel (Web)

- [x] `/admin/hospitals/{id}/inpatient`: occupancy stats, add capacity, admit, active list + discharge, beds-by-ward map
- [x] Nav link from hospital detail
- [x] Edit / delete / deactivate capacity UI
- [x] Maintenance toggle UI
- [x] Bed transfer UI
- [x] Admission history / detail
- [x] Discharge notes field in UI
- [x] Patient picker beyond first page (100) `(search + page size 50)`

Mobile

- [ ] Inpatient / bed board `(out of scope)` unless a clinician mobile surface is planned later Client already has the HTTP methods if needed.

Docs

- [x] `docs/03_Domain_Modules.md`: Ward / Room / Bed / InpatientAdmission
- [x] `docs/05_Database_Design.md`: inpatient tables
- [x] `docs/06_Key_Workflows.md`: admit / discharge
- [x] `docs/00_Change_Log.md`: inpatient entry

Step 2 status

100%. Lifecycle complete on API → Client → Web (capacity, maintenance, transfer, history, discharge notes, patient search). Demo seed + companion docs done. Do not start clinician Mobile for this until product asks for it.

Step 3: Patient Mobile (concept parity + APIs) - 60%

Concept: `C:\laragon\www\raphcare-mobile-app-concept`. Screens and tokens: `docs/Mobile_Concept_Port.md`. Flags: `RaphCare.Mobile.Kernel` / `FeatureFlags`.

Shell, auth, onboarding

- [x] Landing / welcome
- [x] Sign-in (email) + verify email
- [x] Register options: email / phone / voice
- [x] Phone OTP send / verify -> API JWT
- [x] Voice onboarding (record -> API)
- [x] Account created
- [x] Home dashboard + quick links
- [x] Feature-flagged navigation / under-construction fallback

Verticals (API + Client + MAUI)

Area	API	Client	Mobile UI	Notes
Appointments	[x]	[x]	[x]	List / book / detail
Care / telehealth	[x]	[x]	[x]	Request call + join; Agora on Android
Health records	[x]	[x]	[x]	List + detail
Devices / BLE vitals	[x]	[x]	[x]	Offline outbox retries
Insurance	[x]	[x]	[x]	Hub + add / detail
Billing	[x]	[x]	[x]	Hub + add payment method
Family members	[x]	[x]	[x]	List / add / detail
Mental health	[x]	[x]	[x]	Content + mood check-in
AI assistant	[x]	[x]	[x]	Real Azure OpenAI or placeholder reply
Notifications	[x]	[x]	[x]	List / mark read / push registration
Settings / profile	[x]	[x]	[x]	Edit, personal info, medical info, emergency contacts, privacy, help, language, change password

Screen visual parity table is marked done in `Mobile_Concept_Port.md` (last pass). Re-check when the React concept changes.

Mobile device coverage (phones, OS, wearables)

Phones / OS targets:

- [x] Android app build and run path (primary day-to-day target)
- [ ] iOS app build and run path verified on a physical iPhone `(partial)`: project targets iOS; release spot-check still open
- [ ] Same major flows verified on **both** Android and iOS before calling a vertical release-ready (use `Mobile_Release_Ready_Checklist.md` per-screen table)
- [ ] Windows MAUI target `(out of scope for BLE / most device work)`: BLE not active on Windows in this solution
- [ ] Mac Catalyst as a BLE test host `(optional)`: supported for Bluetooth perms; not a patient ship target

Telehealth video on device:

- [x] Android in-call UI + Agora RTC wiring
- [x] API Agora token mint verified (`tools/verify-agora-rtc-config.ps1`)
- [ ] iOS AgoraRtcKit (xcframework) wiring `(blocked / partial)`: see `docs/10_Agora_Twilio_Setup.md`
- [ ] End-to-end video call on a real Android phone `(manual)`: server token OK; needs camera device. See `docs/10_Agora_Twilio_Setup.md` §2.7
- [ ] End-to-end video call on a real iPhone `(blocked: iOS Agora)`

Push notifications on device:

- [x] App can register a push device token with the API
- [ ] Android FCM delivery in a configured environment `(partial)`: needs Firebase service account on API
- [ ] iOS push delivery (APNs / FCM path) verified on a physical iPhone `(partial)`
- [ ] Push off / no-op when Firebase is not configured (expected today)

Wearables / patient hardware (see `docs/11_Devices_BLE_E580_E585.md`, `docs/13_Patient_Device_Packages_and_Fleet.md`, **`docs/14_Wearable_Capability_Catalog.md`**):

- [x] Devices screen: BLE scan, connect, disconnect
- [x] E580 / E585 name filter (`E585E580DeviceFilter`, includes `ET580` / `ET585`) + "show all BLE" fallback
- [x] Android Bluetooth / Nearby devices permission flow
- [x] iOS Bluetooth usage string (`Info.plist`)
- [x] Heart-rate GATT read when the band exposes standard HR (`0x2A37`)
- [x] SpO_2 GATT read when the band exposes standard PLX (`0x2A5F` / `0x2A60`)
- [x] Vitals sync to API + offline outbox retry (`FileVitalsSyncOutbox`): HR / SpO_2 batches only

- [x] SKU-agnostic wearable capability catalog documented (`docs/14_Wearable_Capability_Catalog.md`)
- [x] HBand Android vendor path (JNI): download script + `HBandAndroidWearableBridge` + connect/pwd/person + live HR/SpO₂ start `(partial)`: needs physical ET580/ET585 verification; iOS not wired
- [ ] Reliable full band data (HBand-class parity for in-scope metrics) `(partial)`: Phase 1 HR/SpO₂ hooks in; activity/sleep/history still open. See `docs/14`
- [ ] Full typed HBand SDK C# binding project `(out of scope for Phase 1)`: JNI bridge used instead; optional later. See `docs/12_HBand_SDK_Integration.md`
- [ ] Live HR + SpO₂ in patient app via vendor protocol verified on hardware `(partial)`: wired; awaiting device test
- [ ] Auto sync / background monitoring of band readings `(partial)`: manual sync + outbox exist; background unproven; OS limits in `docs/14`
- [ ] Activity (steps / kcal / distance / goals) domain + sync + mobile `(not started)`
- [ ] Sleep domain + sync + mobile `(not started)`
- [ ] Stress domain + sync + mobile `(not started)`
- [ ] Body temperature domain + sync + mobile `(not started)`
- [ ] ECG / PPG capture + sync `(not started)`: domain `ECGReading` exists; no patient sync / BLE yet
- [ ] Glucose / body composition / Health Glance from band `(not started)`: gate clinical use; see guardrails in `docs/14`
- [ ] Weather / companion pushes to watch `(out of scope for clinical v1)` unless product expands lifestyle parity
- [ ] Y6 Pro (4G SOS / fall / emergency fleet workflow) end-to-end in app `(partial / product path)`: webhook + SMS + clinic admin UI (patient chart + hospital Devices board); OEM mapping / push alerts / full app vertical still open
- [ ] Clinic-facing view of patient device readings beyond admin devices list `(partial)`

Permissions / hardware UX:

- [x] Bluetooth permission prompts documented for Android 12+ and iOS
- [ ] Camera / mic permission path for telehealth verified on both OS `(partial)`
- [ ] Airplane / no-network degraded behavior spot-checked on a real phone
- [ ] Low-battery / Bluetooth-off empty states feel clear on Devices

Mobile cross-cutting left

- [ ] AI assistant production LLM `(partial)`: needs `PatientAssistant` Azure OpenAI config
- [ ] Dark mode resource dictionary `(out of scope for v1)` unless product reverses
- [ ] Deep links / NotFound route UX
- [ ] Prod feature-flag rollout plan filled in `Mobile_Release_Ready_Checklist.md`
- [ ] Accessibility spot-check on phone (labels, contrast, key flows)
- [x] Store packaging: icons, splash, package ids, privacy strings for Play / App Store `(partial)`: Android package id `com.yindula.raphcare`; Azure + Play Internal runbook; **`RaphCare.Web.Host`** for

Linux App Service; Play Console upload and iOS store still manual

Admin panel

- [ ] Patient-app verticals on Web `(out of scope)`: patients use Mobile; staff use admin hospital flows

Step 3 status

60%. Concept screens and patient APIs are largely in. Android is ahead of iOS for video; BLE E580/E585 path exists; wearable catalog is in `docs/14`. Catch-up: phone/OS verification, iOS Agora, push config, store packaging, and wearables depth (live HR/SpO_2, auto sync, activity/sleep and related metrics).

Step 4: Staff / shared clinical APIs (non-admin-clinic) - 83%

Broader API surface used by staff tools or integrations (not the patient app primary path).

- [x] `PatientsController`, `ClinicalController`, `AppointmentsController` (staff-shaped)
- [x] `DevicesController`, `TelemedicineController`, `BillingController`, `InsuranceController`
- [x] `MentalHealthController` (staff assessments: placeholder data until persistence)
- [x] `AIController`, `ReportingController`
- [x] FHIR export (`api/fhir`)
- [x] Standalone emergency webhook
- [x] Clinic admin surfaces emergency events (patient chart + hospital Devices board)
- [ ] Staff mental-health assessments persistence `(partial)`: list exists; real store TBD
- [ ] Reporting depth / real dashboards beyond placeholders `(partial)` as product defines

Step 4 status

83%. Core staff and integration controllers are in. Mental-health persistence and reporting depth still thin.

Cross-cutting - 86%

- [x] Solution layers: Domain -> Application -> Persistence / Infrastructure / Identity -> API; Client -> Web / Mobile
- [x] `RaphCare.Client` + `AddRaphCareClient` for Web and Mobile
- [x] Dual auth: Entra JWT (staff) + patient OTP / email JWT
- [x] MPI / patient merge pipeline (see audit docs)
- [x] Inpatient documented in companion docs (Step 2 docs rows)

- [x] Web UI localization (en / fr / ln / sw): `AppResources` + language picker on auth shell and admin layout
- [ ] E2E smoke script covering admin inpatient + one patient mobile vertical against a running API

Cross-cutting status

86%. Platform wiring includes Web UI languages. Missing an automated E2E smoke against a running API.

Rollup

Area	%	Notes
Step 1 Admin outpatient	100%	Clinician Mobile out of scope
Step 2 Inpatient MVP	100%	Clinician Mobile out of scope
Step 3 Patient Mobile	60%	iOS video, push, wearables depth
Step 4 Staff / shared APIs	83%	Persistence / reporting polish
Cross-cutting	86%	E2E smoke open
Overall (scored items)	**78%**	Out of scope / open optional excluded
Without wearable Phase 2+ metrics	**82%**	Excludes activity / sleep / stress / temp / ECG / glucose rows

Next: iOS Agora + dual-OS smoke, push config, wearable live vitals prove-out, then E2E smoke script.

Quick surface map

Surface	Primary owner	Status snapshot
API admin clinics	`AdminClinicsController`	Outpatient complete; inpatient lifecycle complete
API patient	`api/patient/*`, auth, onboarding	Verticals wired; config-dependent push / AI / iOS RTC
Client	`IAdminClinicService`, patient `I*Service`	Matches current APIs
Web admin	`Pages/Admin/Hospitals/*`	Hospital ops + inpatient lifecycle; UI en/fr/ln/sw; phone use later (responsive + thin install, not a full PWA)
Mobile	`RaphCare.Mobile` patient app	Concept screens in; Android ahead of iOS for video; BLE partial

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Related docs

Doc	Use for
`raphcare-feature-checklist-partner.md`	Plain-language status for non-technical partners
`../Mobile_Concept_Port.md`	Tokens, route map, screen visual parity
`Mobile_Release_Ready_Checklist.md`	Flags, quality bar, release backlog
`../09_Mobile_App_Guide.md`	Mobile structure, DI, config
`../11_Devices_BLE_E580_E585.md`	BLE bands (E580 / E585)
`../12_HBand_SDK_Integration.md`	HBand SDK binding path
`../13_Patient_Device_Packages_and_Fleet.md`	Y6 / E580 / E585 fleet SKUs
`../14_Wearable_Capability_Catalog.md`	SKU-agnostic band features + delivery phases
`../10_Agora_Twilio_Setup.md`	Telehealth RTC setup
`../15_Web_Admin_On_Phone.md`	Staff phone use of web admin: responsive UI, then thin install; not a full PWA
`../02_Solution_Structure.md`	Project layout
`../06_Key_Workflows.md`	Workflow narratives (update when inpatient ships docs)